

**Energy-Efficient Dual Hydroponic System for
Organic/Inorganic Capsicum Chili Farming**

25-26J-035

Project Proposal Report

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BSc (Hons) in Information Technology

Specializing in Computer Systems & Network Engineering

Department of Computer System Engineering

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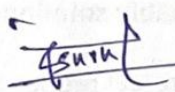
Department of Computer System Engineering

Sri Lanka Institute of Information Technology

September 19

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date: 19/9/25

ABSTRACT

Hydroponic farming has emerged as a sustainable alternative to conventional agriculture, addressing challenges such as soil degradation, limited arable land, rising costs of fertilizer, and climate variability. Among hydroponic methods, the Nutrient Film Technique (NFT) is notable for its efficient water use and continuous nutrient delivery. However, while inorganic NFT systems are well-studied, organic NFT hydroponics remain underdeveloped. Systems using compost tea or similar extracts face nutrient variability, microbial instability, and labor-intensive management, limiting their adoption particularly in regions like Sri Lanka where affordable and sustainable solutions are crucial.

This project proposes the design and implementation of a semi-automated organic NFT hydroponic system tailored for capsicum chili cultivation. The system integrates a network of sensors, including pH, EC, temperature, humidity, water level, gas, and color sensors, to monitor both environmental and compost solution conditions. Sensor fusion techniques will be used to analyze compost maturity by detecting volatile emissions and color variations, classifying the compost into Fresh, Ready, or Over-fermented states. A fuzzy logic-based control algorithm will automate nutrient dosing, irrigation scheduling, and pH balancing to ensure consistent root-zone chemistry. Additionally, a cloud-enabled dashboard will provide farmers with real-time data visualization, alerts, and manual override options, while an air-gap reservoir will enhance root oxygenation and system resilience.

The proposed system aims to minimize farmer intervention, reduce dependency on synthetic fertilizers, and stabilize crop yield. By combining IoT, automation, and intelligent decision-making, this research contributes a practical and scalable solution for sustainable agriculture. It has the potential to empower smallholder farmers, improve resource efficiency, and promote environmentally responsible food production in Sri Lanka and beyond.

Keywords

IoT hydroponics, fuzzy logic, compost tea classification, organic NFT, capsicum chili.

TABLE OF CONTENTS

DECLARATION	i
ABSTRACT	ii
TABLE OF CONTENTS	iii
LIST OF TABLES	v
LIST OF FIGURES	vi
LIST OF ABBRIVIATION	vii
1. INTRODUCTION	1
1.1 Background	2
1.1.1 Challenges in Conventional Agriculture.....	2
1.1.2 Hydroponics as a Sustainable Alternative	3
1.1.3 The Potential and Challenges of Organic NFT Hydroponics	3
1.1.4 Emerging Role of IoT and Intelligent Control.....	4
1.2 Literature Review	5
1.2.1 NFT Hydroponics and Capsicum Nutrient Management.....	5
1.2.2 Organic Nutrient Solutions and Compost Quality	5
1.2.3 IoT Architectures for Real-Time Monitoring and Alerts.....	6
1.2.4 Intelligent Control: From Thresholds to Fuzzy Logic	7
1.2.5 Synthesis: What’s Missing and How Your Method Addresses It.....	7
1.3 Research Gap	8
1.4 Research Problem.....	10
2. OBJECTIVES	11
2.1 Main Objectives	11
2.2 Specific Objectives	11
3. METHODOLOGY	12
3.1 System Diagram	12
3.2 Project Execution Approach.....	14
3.3 Materials and Equipment.....	15
3.4 Work Breakdown Structure (WBS).....	16
3.5 Gantt Chart	18
3.6 Expected Results	18

4. PROJECT REQUIREMENTS	20
4.1 Functional Requirements.....	20
4.2 Non-Functional Requirements.....	21
4.3 Expected Test Cases	21
4.5 Hardware Requirements	22
4.6 Software Requirements	23
5. BUDGET AND BUDGET JUSTIFICATION.....	24
5.1 Budget Table	24
5.2 Budget Justification	25
6.CONCLUSION	26
7. REFERENCE.....	27

LIST OF TABLES

Table 1.1 : Gaps and Proposed System	8
Table 5.1: Budget and Justification	24

LIST OF FIGURES

Figure 3.1: System Diagram	12
Figure 3.4: Gantt Chart	18

LIST OF ABBRIVIATION

Abbreviation	Full Form
NFT	Nutrient Film Technique
IoT	Internet of Things
pH	Potential of Hydrogen
EC	Electrical Conductivity
TDS	Total Dissolved Solids
VOC	Volatile Organic Compounds
ESP32	Espressif Systems 32-bit Microcontroller
MQTT	Message Queuing Telemetry Transport
HTTP	Hypertext Transfer Protocol
LED	Light Emitting Diode
DHT22	Digital Humidity and Temperature Sensor
MDPI	Multidisciplinary Digital Publishing Institute
GUI	Graphical User Interface
AI	Artificial Intelligence
CSV	Comma-Separated Values
Wi-Fi	Wireless Fidelity
LKR	Sri Lankan Rupee

1. INTRODUCTION

This section introduces the function of designing and implementing a semi-automated organic NFT hydroponic system with intelligent monitoring and nutrient management for capsicum chili (*Capsicum annuum*) cultivation.

Hydroponics has emerged as a sustainable alternative to conventional soil-based agriculture by directly supplying plant roots with nutrient solutions, allowing precise control of pH, electrical conductivity (EC), and mineral composition [1], [2]. This soilless approach improves water and fertilizer efficiency while addressing issues of soil degradation, water scarcity, and limited arable land [3], [6]. Among hydroponic techniques, the Nutrient Film Technique (NFT) is particularly suited for high-value crops such as capsicum, where growth and yield are highly sensitive to nutrient balance, environmental stability, and microclimatic control [4], [5]. Recent reviews highlight NFT's adaptability, efficiency, and scalability, making it a key technique for urban and protected cultivation systems [2].

Although inorganic hydroponics is well-studied, organic NFT systems where nutrients are derived from compost teas or organic waste extract remain underexplored and face distinct challenges. These include nutrient variability, pH and EC fluctuations, and microbial instability that reduce consistency in crop performance [1], [3], [19], [20]. Studies on organic nutrient solutions have shown that process parameters such as pH, EC, total dissolved solids (TDS), and ion concentration strongly affect yield and quality, requiring close monitoring and process control [19], [20]. Composting research further suggests that volatile organic compounds (VOCs) and colorimetric changes are reliable indicators of compost maturity and nutrient quality, creating opportunities for sensor-based quality assessment [11], [12].

At the same time, advancements in smart agriculture demonstrate how Internet of Things (IoT) platforms, cloud services, and low-cost microcontrollers (e.g., ESP32, Raspberry Pi) can transform hydroponic management. IoT-enabled hydroponic systems now integrate pH, EC, temperature, humidity, and water-level sensors with real-time dashboards, alerts, and data logging, reducing manual intervention and improving decision-making [6]– [8], [10], [18]. These architectures provide remote

access and automation capabilities, making them viable for smallholders and urban farms.

Control strategies are also evolving from simple threshold-based rules to intelligent approaches. Fuzzy logic has been successfully applied in NFT and other hydroponic systems to regulate irrigation, pH, and nutrient dosing under uncertain and non-linear conditions [9], [13]– [17]. Compared to fixed thresholds, fuzzy logic provides smoother control responses and better adaptation to dynamic plant needs. When combined with IoT monitoring, fuzzy controllers enable semi-automated nutrient management that enhances both crop yield and system reliability [16], [17].

Guided by these developments, this project proposes the design and implementation of a semi-automated organic NFT hydroponic system for capsicum chili cultivation. The system integrates multi-sensor monitoring (pH, EC, VOC, color, temperature, humidity), a compost-quality classification model, and fuzzy logic-based control for nutrient dosing, irrigation, and pH regulation. A cloud dashboard will provide visualization, alerts, and manual override options. By addressing variability in organic nutrients and leveraging IoT-enabled automation, the proposed system aims to optimize crop growth, improve resource use efficiency, and support sustainable capsicum production in Sri Lankan conditions.

1.1 Background

1.1.1 Challenges in Conventional Agriculture

Global agriculture is under increasing pressure due to population growth, urbanization, and changing climatic conditions. Conventional soil-based farming methods, while historically productive, are increasingly unsustainable. Soil degradation from over-cultivation, erosion, and excessive use of agrochemicals has reduced fertility and crop resilience [1], [3]. Weather variability, including droughts and floods, further disrupts production, creating uncertainty for farmers.

A major bottleneck is the heavy reliance on synthetic fertilizers. While these inputs ensure predictable nutrient availability, they also increase production costs and contribute to greenhouse gas emissions and water pollution through nutrient runoff [6]. Farmers in developing nations, including Sri Lanka, are particularly vulnerable: they

face rising fertilizer import costs, limited access to advanced irrigation technologies, and declining profitability from small landholdings [3], [6]. These issues collectively highlight the urgent need for sustainable and resource-efficient agricultural systems that reduce dependence on chemical fertilizers while ensuring food security.

1.1.2 Hydroponics as a Sustainable Alternative

Hydroponics has emerged as a promising technology that eliminates soil and instead delivers nutrients directly to plant roots in water-based solutions [1], [2]. This system allows precise management of pH, electrical conductivity (EC), and nutrient concentrations, enabling higher yields with significantly less water and land [1], [3]. For instance, studies show that hydroponic methods can reduce water consumption by up to 90% compared with conventional agriculture [2].

Among hydroponic techniques, the Nutrient Film Technique (NFT) stands out for its efficiency and scalability. NFT systems continuously circulate a thin film of nutrient solution across plant roots, ensuring optimal hydration and aeration while minimizing water use [2], [4]. Capsicum chili (*Capsicum annuum*), one of Sri Lanka's most commercially important vegetable crops, is highly responsive to nutrient balance and microclimatic conditions [4], [5]. NFT is well-suited for chili cultivation because of its ability to maintain consistent nutrient delivery and root oxygenation.

Despite these advantages, most NFT hydroponic systems rely on inorganic nutrient solutions. These are chemically stable and easier to automate but come with limitations: high recurring costs, environmental risks from runoff, and dependence on chemical inputs [2], [6]. This dependency restricts adoption among smallholder farmers who need cost-effective, environmentally sustainable methods that align with local resource availability.

1.1.3 The Potential and Challenges of Organic NFT Hydroponics

Organic hydroponics, or bioponics, is gaining attention as a sustainable evolution of NFT systems. Instead of synthetic salts, nutrients are supplied through compost teas, food waste extracts, or manure-based formulations [1], [3], [19], [20]. These organic inputs provide a circular economic solution, turning waste into resources and reducing

reliance on costly chemical fertilizers. Consumer demand for pesticide- and chemical-free produce also makes organic hydroponics attractive in both local and export markets.

However, several challenges limit the wider adoption of organic NFT systems. Unlike inorganic solutions, organic compost teas exhibit high variability in chemical and microbial composition, leading to unstable pH and EC values [19], [20]. Nutrient concentrations fluctuate depending on compost source, maturity, and environmental factors, which directly affect plant nutrient uptake and yield [19]. Studies on lettuce and other vegetables show that while organic systems can produce competitive yields, process parameters such as TDS, EC, and pH must be closely monitored and controlled to avoid deficiencies or toxicities [19], [20].

Current organic NFT setups rely on manual nutrient management methods such as handheld pH/EC meters or laboratory testing [5]. These approaches are time-consuming, labor-intensive, and often fail to capture the real-time dynamics of microbial activity in compost tea. Recent compost science suggests that volatile organic compounds (VOCs) and colorimetric indicators are reliable markers of compost maturity [11], [12]. Yet, no NFT hydroponic system currently integrates these parameters into automated control frameworks. This creates operational uncertainty, reduces reliability, and limits adoption in commercial or smallholder contexts.

For Sri Lanka, where smallholder farmers could benefit greatly from low-cost and sustainable alternatives, the lack of intelligent, automated organic NFT solutions represents a major technological gap.

1.1.4 Emerging Role of IoT and Intelligent Control

IoT technologies now enable real-time hydroponic monitoring through sensors (pH, EC, temperature, humidity, water level) linked with microcontrollers such as ESP32 and cloud dashboards [6]– [8], [10], [18]. These systems reduce manual effort and support automation, but most are designed for stable inorganic solutions [6], [7], [8].

Fuzzy logic has also shown strong potential for regulating irrigation, pH, and nutrient dosing under uncertain conditions [9], [13]– [17]. Studies applying fuzzy logic in NFT systems demonstrate more stable control than threshold methods [16], [17]. Yet, no

framework combines IoT monitoring, compost-quality classification, and fuzzy logic for organic NFT hydroponics. Developing such a system could reduce farmer dependency, improve yield stability, and enhance sustainable chili production in Sri Lanka.

1.2 Literature Review

1.2.1 NFT Hydroponics and Capsicum Nutrient Management

Palmitessa et al. synthesize nearly five decades of NFT research and position NFT as a high-efficiency, scalable technique for controlled environments, but largely within inorganic regimes [2]. For capsicum specifically, Yeo et al. developed stage-wise nutrient compositions for paprika (*Capsicum annuum*), demonstrating that growth and yield are highly sensitive to EC/pH targets that shift across phenological stages [4]. Ding et al. quantified how nutrient concentration deviates can induce abiotic stress in sweet pepper directly linking over/under-supply to reductions in photosynthetic performance and biomass [5].

Implications. Your controller should support stage-dependent setpoints (seedling/vegetative/flowering/fruitlet) and treat deviations in EC/pH as potential stressors, not just setpoint errors.

Limitations- These references assume stable, mineral solutions. They do not address the temporal variability found in organic extracts, nor do they propose online composition sensing.

1.2.2 Organic Nutrient Solutions and Compost Quality

Park reviews “organic hydroponics” (biaponics), noting strong sustainability potential but also variability in nutrient release and microbial dynamics that complicate control [1]. Szekely et al. argue biaponics can be viable but emphasize process monitoring needs to stabilize outcomes [3]. Two experimental papers are particularly valuable for your case: Dalogdog et al. test organic nutrient solutions and show that pH, EC, TDS, and ions correlate with quality and yield, underscoring the need for process quality control [19]; Olmedo et al. extend this, proposing organic solutions as inputs for process control and highlighting where control points should be placed (make-up water, acid/base dosing, and dilution/flush events) [20].

In parallel, compost science shows how volatile organic compounds (VOCs) and colorimetric indicators act as maturity proxies: Berger et al. present a two-component sampling strategy for VOCs relevant to compost stages [11]; Szyłak-Szydlowski et al. quantify odour/VOC emissions across composting phases and relate them to process conditions [12].

Implications. These works justify a sensor-fusion classifier (gas + color) that labels compost tea as Fresh / Ready / Over-fermented and feeds that state into dosing logic.

Limitations- Organic studies often stop at correlation, not closed-loop automation; compost papers validate signals, not hydroponic dosing decisions. No one closes the loop from compost maturity to nutrient dosing in NFT.

1.2.3 IoT Architectures for Real-Time Monitoring and Alerts

Shin et al. present a low-cost IoT hydroponic platform (open-hardware ESP32, pumps/relays, pH/EC/temperature, mobile dashboard), proving affordability and feasibility for smallholders [6]. Febriana et al. show IoT-based monitoring + control pipelines that improve responsiveness and remote oversight in web-centric deployments [7]. Sneineh et al. detail an ESP32 + cloud architecture for hydroponic monitoring, covering sensor integration, telemetry, and dashboarding [8]. Awal et al. emphasize real-time monitoring and cloud integration for small-scale producers [10]. From your uploaded set, Gourshettiwar & Reddy contribute a monitoring + alert model with explicit threshold triggers, illustrating how on-device logic and cloud notifications can be combined [18]. Technologies used- ESP32 microcontrollers; pH/EC/temperature/humidity/level sensors; cloud dashboards (mobile/web); rule-based triggers; sometimes Blynk/REST/MQTT for telemetry. Results- Robust remote visibility, faster fault detection, and reduced manual checks; working cost points are compatible with resource-constrained settings.

Limitations- (i) Nearly all operate with inorganic feeds; (ii) control is often threshold-based; (iii) no compost sensing (VOCs/color) is integrated; (iv) reliability under network/power interruptions and sensor drift is rarely characterized.

1.2.4 Intelligent Control: From Thresholds to Fuzzy Logic

Fuzzy logic has proven effective in hydroponics where plant–solution dynamics are uncertain. Rivai et al. showed feasibility of on-device fuzzy inference in NFT within an IoT setting [13]. Puno et al. designed an NFT controller using pH, EC, and water-level inputs with a 27-rule base for nutrient and water dosing [16]. Triantino et al. demonstrated smoother pH regulation than thresholds [17], while Olalla et al. improved irrigation stability in strawberries [9]. Kumar [14] and Rosli [15] confirmed fuzzy logic’s adaptability to crop-specific water and nutrient management.

Technologies- Mamdani inference; inputs: pH, EC, level/flow, temperature; outputs: dosing pumps/valves for nutrients, water, acid/base; implementation via ESP32/Arduino. Results- Reduced oscillations, faster settling, and more stable EC/pH.

Limitations- Mostly designed for mineral solutions; little attention to compost variability, sensor health, or fail-safes. Outputs are often binary on/off without rate control, and systems lack transparency for farmer trust.

1.2.5 Synthesis: What’s Missing and How Your Method Addresses It

Gaps. (i) Compost-quality signals (VOCs, color) are validated [11], [12] but not linked to dosing. (ii) Organic NFT work highlights correlations [19], [20] but lacks automation. (iii) IoT systems monitor inorganic feeds [6]– [8], [10], [18] but ignore organic variability. (iv) Fuzzy controllers regulate pH/EC [9], [13]– [17] but are not tuned for compost state or capsicum stage [4], [5].

Methodology response.

- **Sensors:** Add gas + color sensing; classify compost tea as Fresh/Ready/Over-fermented [11], [12].
- **Control:** Apply fuzzy logic conditioned on compost state and capsicum growth stage [4], [5], [9], [13]– [17].
- **Process quality:** Include dilution/flush cycles, micro-dosing, and safe mixing based on [19], [20].
- **Reliability:** Use ESP32/cloud IoT [6]– [8], [10], [18] with fail-safes, buffering, and sensor-drift checks.

1.3 Research Gap

- Research Gap Comparison Table

Table 1.1: Gaps and Proposed System

Features	[1] Park (2024)	[2] Palmitessa et al. (2024)	[3] Szekely et al. (2022)	[16] Puno et al. (2020)	[19] Dalogdo et al. (2023)	Proposed System
Focus on Organic Nutrients	✓	✗	✓	✗	✓	✓
Real-time Compost Classification	✗	✗	✗	✗	✗	✓
Intelligent Nutrient Automation (Fuzzy Logic)	✗	✗	✗	✓	✗	✓
IoT-enabled Dashboard with Compost Monitoring	✗	✗	✗	✗	✗	✓
Reduced Farmer Dependency / Semi-Automation	✗	✗	✗	✓	✗	✓

- Existing NFT hydroponic systems primarily focus on inorganic nutrients

Most NFT-based hydroponic systems still depend on synthetic nutrient solutions because of their predictable mineral composition and ease of automation [2], [6]. Studies have demonstrated IoT-based real-time monitoring and fuzzy logic for nutrient and pH control in these systems [6], [9]. However, these models overlook organic nutrient sources such as compost tea, which are increasingly favored due to sustainability, rising fertilizer costs, and the demand for chemical-free produce [1], [3]. This lack of organic integration limits adoption in regions like Sri Lanka, where locally sourced organic inputs are more economical and environmentally sustainable than synthetic fertilizers.

- Organic NFT systems remain largely manual.

Studies on biaponics confirm that compost-based nutrients can support crop growth [3], but current systems rely on manual observation of pH/EC values and farmer judgment of compost maturity [19]. This approach is inconsistent, labor-intensive, and error-prone, especially for sensitive crops like capsicum chili. Without semi-automated regulation, organic systems cannot deliver the yield stability demonstrated in inorganic NFT setups.

- No real-time compost classification integrated into dosing.

Existing organic hydroponics typically depend on thresholds like EC or TDS, which do not capture the biological maturity or chemical quality of compost tea [19]. Compost science shows that volatile organic compounds (VOCs) and color indicators strongly correlate with fermentation stages [11], [12]. However, no NFT system fuses these signals into a real-time classification model (e.g., Fresh, Ready, Over-fermented) that informs nutrient dosing. This gap leaves nutrient delivery uncertain and unoptimized.

- IoT dashboards neglect compost dynamics.

Most IoT-enabled hydroponic systems focus on inorganic nutrient monitoring tracking pH, EC, temperature, and humidity [6], [7], [8], [10]. While these platforms provide visualization and alerts, they do not address the dynamic variability of compost tea, such as microbial activity or maturity stage. Without compost integration, dashboards cannot guide organic nutrient decisions. A platform that unites compost classification, fuzzy-logic dosing, and manual override would reduce farmer dependency and improve trust in organic NFT systems.

1.4 Research Problem

Organic hydroponic farming using the Nutrient Film Technique (NFT) offers a sustainable alternative to chemical-based agriculture, yet its adoption remains limited due to the unpredictability of compost tea nutrient composition, absence of real-time quality monitoring, and lack of automated nutrient regulation mechanisms. Existing NFT systems in Sri Lanka are either inorganic-based or rely on manual testing of compost tea, which often leads to inconsistent nutrient availability, pH imbalances, and yield fluctuations in capsicum chili cultivation. Furthermore, most available smart farming platforms emphasize sensor monitoring without integrating intelligent decision-making to adapt dosing and irrigation schedules based on compost maturity or environmental changes. This gap hinders scalability, increases labor dependency, and reduces efficiency, especially in resource-constrained and off-grid farming environments.

Therefore, **Research Questions:**

- How can a multi-sensor network (pH, EC, temperature, humidity, water level, VOC, and color) be integrated and calibrated with an ESP32 microcontroller to enable real-time monitoring of compost-nutrient and environmental conditions?
- How can sensor fusion of VOC and color data be used to classify compost tea quality into Fresh, Ready, and Over-fermented stages to guide nutrient dosing strategies?
- How can fuzzy logic assisted algorithms be designed to regulate nutrient dosing, irrigation scheduling, and pH balancing in organic NFT hydroponics for capsicum chili cultivation?
- How can a cloud-based dashboard be designed to visualize sensor data, compost maturity, and system control activities in real time, while providing alerts, analytics, and manual override options for farmers.

2. OBJECTIVES

2.1 Main Objectives

To design and implement a semi-automated organic NFT hydroponic system, enabled with IoT-based real-time monitoring, compost-tea quality classification, and fuzzy-logic assisted control of nutrient dosing, irrigation, and pH balancing, to optimize capsicum chili growth, improve yield stability, and support farmer decision-making with reduced manual effort under Sri Lankan farming conditions.

2.2 Specific Objectives

- To integrate and calibrate a multi-sensor network (pH, EC, temperature, humidity, water level, VOC, and color sensors) with an ESP32 microcontroller for real-time monitoring of compost-nutrient and environmental conditions.
- To develop a compost-tea quality classification model using sensor fusion of VOC and color data to distinguish Fresh, Ready, and Over-fermented compost stages for data-driven nutrient dosing.
- To implement fuzzy logic-assisted control algorithms for automated nutrient dosing, irrigation scheduling, and pH balancing, tuned to the growth requirements of capsicum chili.
- To design and deploy a cloud-based dashboard that visualizes sensor data, compost status, and control activities in real time, while providing alerts, analytics, and manual override options to support farmer decision-making.

3. METHODOLOGY

3.1 System Diagram

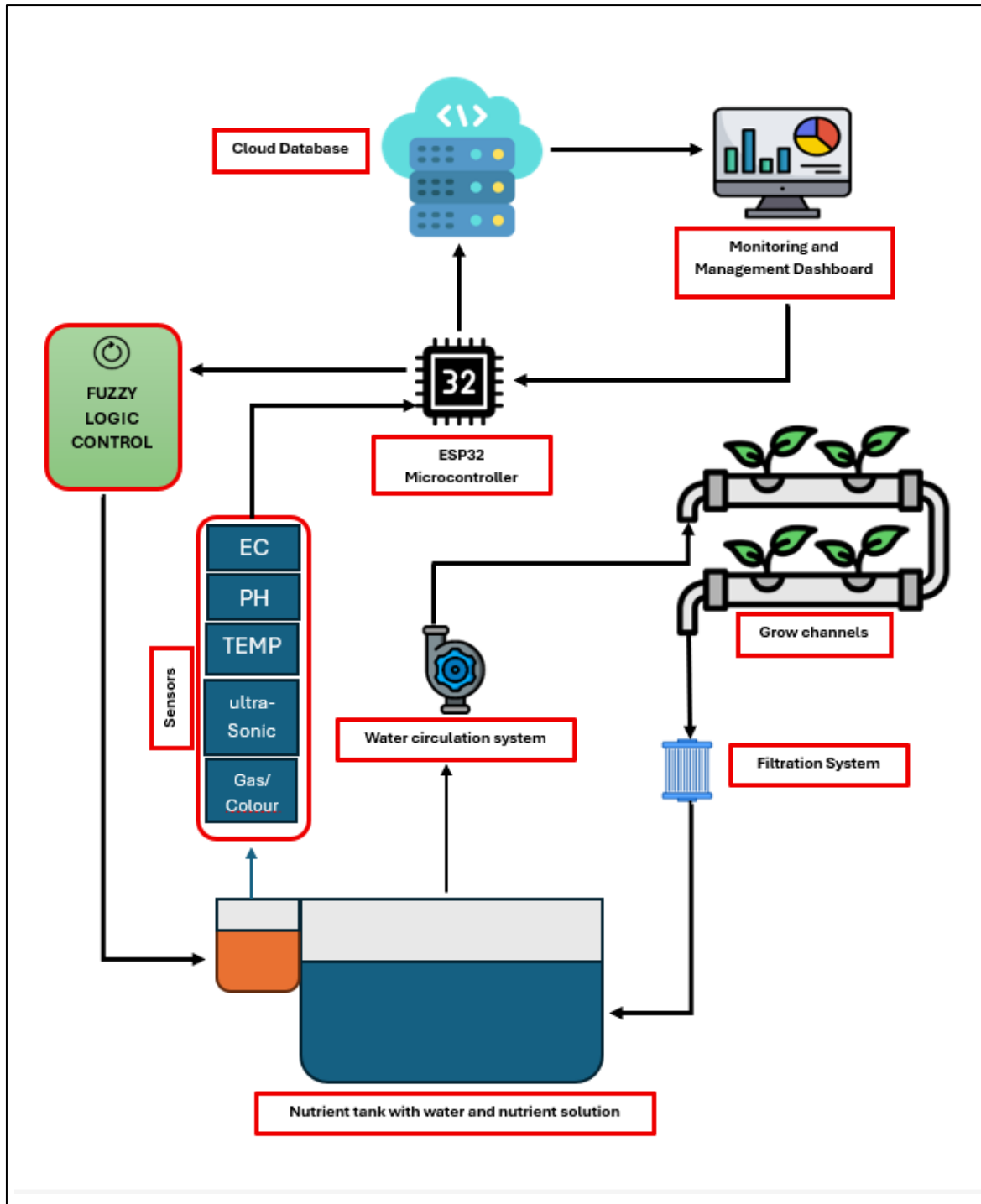


Figure 3.1: System Diagram

The proposed system consists of several interconnected components (Figure 3.1) that work together to achieve semi-automated compost-based nutrient management for capsicum chili growth:

- **Sensors and Data Acquisition:** A combination of pH, EC, TDS, temperature-humidity, ultrasonic water level, gas, and color sensors are deployed to collect continuous data on nutrient solution quality, compost tea maturity, and environmental conditions inside the hydroponic setup. These inputs ensure accurate real-time monitoring of all critical growth factors.
- **Processing and Control (ESP32 Microcontroller):** Acting as the central processing unit, the ESP32 gathers sensor data and applies a fuzzy logic algorithm to classify compost tea states (Fresh, Ready, Over-fermented). Based on this classification, it determines the appropriate adjustments for nutrient dosing, irrigation, and pH balance
- **Actuation and Hydroponic Flow:** Peristaltic pumps, solenoid valves, and an air pump execute the controller's decisions. Pumps deliver compost tea and pH correction solutions, solenoids manage water flow in the NFT channels, and the air pump maintains oxygenation of the root zone via an air-gap reservoir.
- **Cloud Dashboard and User Interaction:** Processed data is transmitted over Wi-Fi to a cloud-based dashboard. This provides real-time visualization, historical analysis, and alert notifications. Farmers can also use manual override controls, ensuring the system is semi-automated leveraging intelligent automation while preserving human decision-making authority.

Together, these components create a streamlined flow: sensors gather data → ESP32 analyzes and decides → actuators respond → cloud dashboard informs and allows user input. This integration enables efficient nutrient management while ensuring adaptability to real-world farming conditions in Sri Lanka.

3.2 Project Execution Approach

The project adopts a hybrid execution approach that blends incremental prototyping with test-driven validation to ensure both functional accuracy and practical usability. Each stage builds upon the previous one, reducing risks and enabling continuous refinement.

- **Design Phase:** The overall system architecture is planned, including nutrient film setup, compost tea integration, and sensor-actuator placement. Fuzzy logic rules for compost classification and nutrient dosing are defined based on literature and expert consultation. A nutrient flow model is prepared to estimate pH, EC, and oxygenation requirements for capsicum chili growth at different stages.
- **Prototype Development:** Hardware components are assembled and tested individually. The ESP32 microcontroller is programmed for basic sensor data acquisition and actuator control. Each sensor (pH, EC, TDS, gas, color, etc.) is calibrated in a lab environment to establish reliable baselines before integration.
- **System Integration:** The sensors and actuators are connected to the ESP32 and tested as a unified system. Fuzzy logic algorithms are implemented to classify compost maturity (Fresh, Ready, Over-fermented) and automate irrigation, nutrient dosing, and pH balancing decisions. The air-gap reservoir and aeration system are also integrated to ensure root oxygenation.
- **Cloud Connectivity:** Data communication protocols are set up to transmit readings from ESP32 to a cloud server. A web-based dashboard is developed to visualize parameters (pH, EC, compost status, climate data) and provide alert notifications. Farmers can access the dashboard via PC or mobile device, with manual override features for pumps and valves.

- **Pilot Testing:** The semi-automated NFT system is deployed with capsicum chili plants under controlled greenhouse conditions. Compost tea is prepared and fed into the system, while sensor data is logged continuously. The system's ability to classify compost tea states, regulate nutrient dosing, and stabilize growth parameters is assessed. Crop growth metrics such as leaf area, plant height, and fruiting are recorded.
- **Optimization:** Based on pilot testing, fuzzy logic rules and pump scheduling are fine-tuned to improve efficiency. For example, dosing intervals may be adjusted to avoid over-irrigation, and aeration cycles optimized to reduce energy consumption. Redundancy checks (e.g., backup thresholds for EC and pH) are also introduced for reliability.
- **Final Evaluation:** The performance of the semi-automated system is compared against a conventional inorganic NFT hydroponic setup. Key evaluation metrics include yield (kg per plant), nutrient usage efficiency, energy consumption, and labor reduction. The comparison highlights the practical benefits of integrating compost-based nutrient management with IoT and fuzzy logic in Sri Lankan farming conditions.

3.3 Materials and Equipment

The system requires both hardware and software resources:

- **Hardware:** ESP32, pH sensor, EC sensor, TDS sensor, DHT22 sensor, ultrasonic water level sensor, MQ-series gas sensor, color sensor, air pump, peristaltic pumps, solenoid valves, air-gap reservoirs, power supply, nutrient film channels, and capsicum seedlings.

- **Software:** Arduino IDE for firmware, fuzzy logic engine implementation in C++, UDP protocol for IoT data, Firebase for database, and Realtime dashboard for visualization.
- **Compost Material:** Locally prepared organic compost tea from plant and kitchen waste, fermented under control conditions. This provides a sustainable and low-cost nutrient alternative, reducing reliance on synthetic fertilizers.
- **Nutrient Additives:** Lime or mild acidic solutions for pH regulation, along with natural growth boosters (e.g., seaweed extract or humic acids) to enhance plant resilience and fruit yield.

This carefully selected combination of low-cost IoT hardware, open-source software, and locally available organic resources ensures the system is cost-effective, scalable, and environmentally sustainable, while being well-suited to Sri Lankan farming conditions.

3.4 Work Breakdown Structure (WBS)

The project will follow a structured timeline beginning on July 24, 2025, and concluding on May 20, 2026, ensuring efficient progress and timely delivery of all milestones. The timeline is divided into five main phases:

- **Phase 1: Project Initiation & System Design** (July – September 2025):

This phase includes brainstorming and requirement analysis, group registration, and submission of the project charter. A detailed literature review will be conducted, followed by drafting and finalizing the system architecture and proposal report.

- **Phase 2: Core Implementation & Sensor Integration** (September – November 2025):

Focuses on sensor selection and procurement, hardware setup with ESP32, sensors, and pumps, and calibration of all units (pH, EC, temperature, humidity, water level, VOC, and color). Initial firmware development and preliminary dashboard setup will also be completed.

- **Phase 3: Intelligent Control & Automation** (November – December 2025):

Development of the compost classification module using VOC and color sensors, and fuzzy logic rule-base design for nutrient dosing, irrigation, and pH balancing. Firmware testing and IoT data connectivity will also be implemented to support automation.

- **Phase 4: Advanced System Features** (January – March 2026):

Integration of the cloud database (Firebase) and completion of the real-time monitoring dashboard with alerts and manual override. Semi-automated control testing, energy efficiency improvements, and system refinements will be performed.

- **Phase 5: Evaluation, Testing & Documentation** (March – May 2026):

Final system testing with capsicum plants will be conducted, including stress testing, optimization, and performance evaluation against conventional NFT. The phase concludes with preparation of the final report, project documentation, and viva presentation.

The accompanying Gantt chart provides a visual representation of the timeline, ensuring clarity on task durations, overlaps, and project milestones.

3.5 Gantt Chart

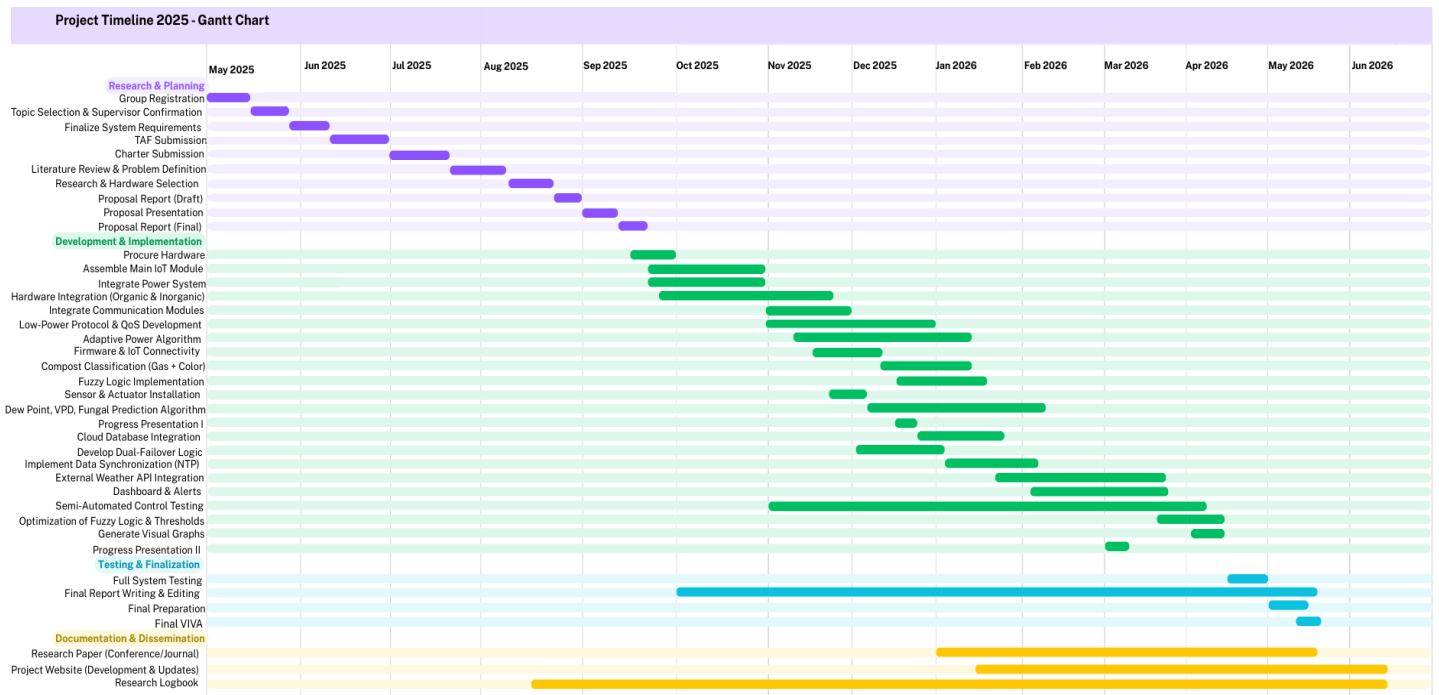


Figure 3.4: Gantt Chart

3.6 Expected Results

The successful implementation of this project is expected to deliver the following outcomes:

1. Stable and Sustainable Crop Growth

- Achieve consistent growth and yield of *Capsicum annuum* (chili) under an organic NFT hydroponic setup.
- Maintain optimal nutrient conditions with reduced variability compared to manual dosing systems.

2. Accurate Compost Tea Classification in Real Time

- Classification of compost tea quality into Fresh, Ready, and Over-Fermented states using a sensor fusion approach (gas + color sensors).

- Expected classification accuracy: $\geq 90\%$ based on sensor calibration and fuzzy rule optimization.

3. Semi-Automated Nutrient and pH Management

- Intelligent dosing of organic nutrient solutions and pH adjustments using fuzzy logic algorithms, reducing manual intervention by at least 50% compared to traditional organic NFT systems.

4. IoT-Enabled Real-Time Monitoring Dashboard

- A cloud-connected dashboard capable of real-time visualization, alerts, and manual override functionality for farmers.
- Continuous logging of key parameters (pH, EC, TDS, temperature, humidity, water level, compost quality).

5. Energy-Efficient System Operation

- Optimized pump scheduling and automation to minimize power consumption while maintaining required nutrient circulation and oxygenation.
- Expected reduction in operational energy by 10–15% compared to fully manual systems.

6. Improved Water and Nutrient Use Efficiency

- Reduction in water wastage through closed-loop NFT circulation and controlled nutrient dosing.
- Anticipated water-use efficiency: $> 90\%$, supporting sustainable farming practices.

7. Comprehensive Performance Evaluation

- Comparative analysis of crop performance between the proposed system and conventional NFT systems, demonstrating better yield stability, lower operational cost, and higher sustainability for smallholder farmers in Sri Lanka

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- Real-time sensing of compost and environmental parameters
 - Continuous monitoring of pH, EC, temperature, humidity, water level, gas emissions, and color changes using calibrated sensors integrated with an ESP32 microcontroller.
 - Enables instant anomaly detection such as pH drift or sudden compost degradation, reducing risk of crop stress.
- Compost classification using gas and color sensors
 - Sensor fusion will analyze VOC concentrations (via gas sensors) and colorimetric indicators (via RGB/color sensors) to identify compost tea states: Fresh, Ready, or Over-fermented.
 - This classification ensures adaptive nutrient management, aligning nutrient delivery with the maturity of compost extracts.
- Semi-automated nutrient dosing and pH balancing via fuzzy logic
 - A fuzzy-logic control engine will process sensor data and suggest/control pump activation for nutrient dosing, water dilution, and pH adjustment.
 - The system will remain semi-automated, allowing manual overrides by farmers for flexibility and fail-safe operation.
- Cloud-based dashboard with alerts and manual override
 - Centralized platform displaying real-time sensor readings, compost quality state, dosing activity logs, and historical analytics.
 - Provides mobile/PC access with SMS/email alerts for critical issues (e.g., over-fermented compost, low water level).

- Includes manual control switches to override automation in case of emergencies.

4.2 Non-Functional Requirements

- Energy efficiency through optimized scheduling
 - Power-aware scheduling of pumps, sensors, and data transmission to maximize use of solar energy and extend battery life.
 - Sleep modes for sensors and adaptive sampling based on compost or environmental stability.
- Scalability for larger crop volumes
 - Modular design allows expansion from a single NFT channel system to multi-channel farms, with additional sensor nodes and dashboards.
 - Supports diverse crops beyond capsicum chili, enhancing real-world applicability.
- User-friendly dashboard interface
 - Clean, multilingual UI (Sinhala/English) with intuitive graphs, color-coded compost states, and alert notifications.
 - Designed to be farmer-friendly, reducing technical complexity and training requirements.

4.3 Expected Test Cases

- **Compost classification accuracy test**
 - Validate model predictions of compost states against laboratory measurements of nutrient concentration and microbial activity.
 - Evaluate accuracy $\geq 85\%$ for practical deployment.

- **Sensor calibration and reliability test**

- Long-duration test of sensors (pH, EC, gas, color) under varying temperature and humidity.
- Assess drift, response time, and re-calibration needs.

- **Fuzzy logic control performance under varying conditions**

- Test system response when compost transitions from Fresh > Ready > Over-fermented.
- Verify stability in maintaining pH (5.8-6.5) and EC (1.8-2.8 mS/cm) without overshoot.

- **System stress test with fluctuating nutrient and environmental changes**

- Simulate sudden disturbances such as power drop, compost imbalance, or water shortage.
- Evaluate system's ability to recover via alerts, overrides, or adjusted control rules.

4.5 Hardware Requirements

The system requires IoT-enabled sensing and control hardware:

- ESP32 Microcontroller – core processing and Wi-Fi communication unit.
- pH Sensor – monitors acidity/alkalinity of the nutrient solution.
- EC Sensor – measures electrical conductivity to track nutrient concentration.
- Temperature & Humidity Sensor (SHT31) – monitors climatic conditions in the greenhouse.
- Ultrasonic Water Level Sensor – monitors nutrient reservoir levels.
- TDS Sensor – measures total dissolved solids in compost tea.
- Gas Sensor (MQ series, e.g., MQ-135) – detects volatile organic compounds (VOCs) for compost quality.

- Color Sensor (TCS3200 or AS7262) – measures compost tea color changes for classification.
- Peristaltic Pumps / Solenoid Valves – for nutrient dosing, irrigation, and pH adjustment.
- Air Pump & Air-Gap Reservoir – ensures oxygenation of plant roots.
- Relays & Power Modules – for actuator control.
- Cloud Storage Module (via ESP32 Wi-Fi) – data transmission to server.

4.6 Software Requirements

The software stack ensures seamless integration of sensing, control, and cloud monitoring:

- **Embedded Programming Environment** – Arduino IDE for ESP32 firmware development.
- **Programming Languages** – C++/Arduino for ESP32 for backend logic if needed.
- **Communication Protocols** – UDP protocol for lightweight IoT data transfer, HTTP/REST for database updates.
- **Database** – Firebase for storing sensor readings, compost classification logs, and historical trends.
- **Cloud Platform** – Google Firebase for IoT integration and visualization.
- **Dashboard Application** – A cloud-connected web dashboard for live visualization of pH, EC, water level, temperature, and plant growth metrics, including manual override controls
- **Data Analytics** – Fuzzy logic engine implemented in firmware for nutrient/pH decision-making.
- **Mobile Access** – Optional mobile-friendly interface for farmers to monitor and control remotely.

5. BUDGET AND BUDGET JUSTIFICATION

5.1 Budget Table

Table 5.1: Budget and Justification

Component	Quantity	Estimated Cost (LKR)	Justification
ESP32 Microcontroller	1	1,500	Acts as the central processing unit of the system, enabling sensor integration, fuzzy-logic control, and IoT communication with the cloud dashboard.
pH Sensor (Liquid Probe)	1	4,500	Measures nutrient acidity/alkalinity , essential for maintaining capsicum chili's optimal root-zone pH (5.8–6.5).
EC Sensor (Electrical Conductivity)	1	8,500	Determines nutrient concentration levels , ensuring balanced compost tea delivery and preventing under/overfeeding.
Temperature & Humidity Sensor (DHT22)	1	3,000	Tracks microclimate conditions , as capsicum growth is highly sensitive to temperature (20–28 °C) and humidity (60–80%).
Ultrasonic Water Level Sensor	1	2,500	Monitors reservoir water levels to prevent pump dry-run and to automate refill alerts.
TDS Sensor	1	4,000	Provides additional measure of total dissolved solids in compost tea, complementing EC readings for improved nutrient assessment.
Gas Sensor (MQ-series, e.g., MQ-135)	1	2,800	Detects volatile organic compounds (VOCs) from compost tea, a key parameter in classifying compost maturity (Fresh/Ready/Over-fermented).
Color Sensor (TCS34725 or equivalent)	1	3,500	Analyzes colorimetric changes in compost tea, aiding sensor fusion with gas sensor for compost quality classification.
Water Pump (12V, submersible)	1	3,200	Provides circulation of nutrient solutions in the NFT channel and supports semi-automated dosing.
Air Pump + Air Stone	1	2,500	Maintains dissolved oxygen levels in the air-gap reservoir, reducing root hypoxia and improving plant health.
NFT Channels (PVC-based)	2	6,000	Channels for plant roots in NFT design, allowing thin film of compost solution to flow consistently.

Reservoir Tank (30L capacity)	1	5,000	Stores compost tea solution and water, integrated with pumps and sensors.
Miscellaneous (Wires, PCB, Connectors, Power Supply, Enclosure)	-	5,500	Supports hardware assembly, protective housing, and reliable electrical connections.

- Total Estimated Cost: $\approx$ LKR 53,000 – 58,000

5.2 Budget Justification

The proposed system requires a combination of sensing, automation, and IoT hardware to achieve semi-automated management of an organic NFT hydroponic system.

- Core Processing & IoT (ESP32): Chosen for its low power consumption, built-in Wi-Fi/Bluetooth, and ability to handle multiple sensors while supporting fuzzy logic and cloud communication.
- Nutrient & Compost Quality Sensors (pH, EC, TDS, Gas, Color): These provide the backbone of nutrient monitoring and compost classification, enabling real-time detection of compost tea quality and solution balance.
- Environmental Monitoring (DHT22): Ensures that the system captures microclimatic variations, since capsicum chili growth is highly sensitive to temperature and humidity stress.
- Automation Components (Water & Air Pumps, Ultrasonic Sensor): These support semi-automation by enabling nutrient circulation, oxygen enrichment, and reservoir level management with minimal farmer intervention.
- Structural Components (NFT Channels, Reservoir Tank, Miscellaneous): Provide the physical infrastructure for hydroponic growth, ensuring that the nutrient film is properly delivered to plant roots while maintaining durability and scalability.

This budget has been designed to strike a balance between cost-effectiveness and functionality, ensuring the prototype is affordable, scalable, and replicable in Sri Lankan farming environments.

6.CONCLUSION

This project successfully introduces a semi-automated organic NFT hydroponic system for capsicum chili cultivation, addressing critical gaps in sustainable agriculture and intelligent nutrient management. Unlike conventional systems that rely on inorganic solutions, the proposed model integrates compost tea as an organic nutrient source, supported by real-time monitoring and intelligent control mechanisms. By combining sensor fusion (pH, EC, TDS, temperature-humidity, VOC, and color analysis) with a fuzzy logic-based control engine, the system ensures precise dosing, pH stabilization, and water reuse while maintaining energy efficiency.

The implementation of IoT connectivity and a cloud-based dashboard enhances transparency and user control, enabling farmers to monitor compost quality, environmental conditions, and nutrient adjustments remotely. The introduction of real-time compost classification categorizing states as Fresh, Ready, or Over-fermented marks a significant innovation, reducing reliance on manual testing and improving decision-making for nutrient management. This advancement directly addresses variability in organic solutions, a primary barrier to the scalability of bioponic systems.

Experimental validation is expected to demonstrate improvements in resource efficiency, crop yield stability, and system reliability compared to conventional NFT setups. Moreover, the modular design ensures scalability for diverse crops and adaptability to local farming conditions, particularly in regions like Sri Lanka, where low-cost, sustainable practices are crucial.

In conclusion, this research bridges the gap between sustainability and automation in hydroponics, offering a cost-effective, farmer-friendly solution that promotes environmental stewardship and food security. Future work can explore AI-based predictive analytics, advanced sensing techniques, and renewable energy integration to further optimize performance and sustainability.

7. REFERENCE

- [1] Y. Park, “Organic hydroponics: A review,” *Scientia Horticulturae*, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0304423823007720>
- [2] O. D. Palmitessa, G. Colucci, and F. Santoro, “Advancements and future perspectives in nutrient film technique (NFT) hydroponics: A bibliometric review (1977–2023),” *Plants*, vol. 13, no. 3, p. 450, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11688204/>
- [3] I. Szekely, A. Tóth, and T. Baranyai, “Bioponics as a promising approach to sustainable hydroponics,” *Water*, vol. 14, no. 23, p. 3975, 2022. [Online]. Available: <https://www.mdpi.com/2073-4441/14/23/3975>
- [4] K. H. Yeo, S. Park, Y. J. Kim, and H. S. Kim, “Development of nutrient solution compositions for paprika (*Capsicum annuum* L.) according to growth stages,” *Horticulturae*, vol. 9, no. 4, p. 412, 2023. [Online]. Available: <https://www.mdpi.com/2223-7747/11/8/1098>
- [5] X. Ding, X. Guo, and J. Sun, “Nutrient concentrations induce abiotic stresses to sweet pepper: Regulation of growth and physiology,” *Plants*, vol. 11, no. 8, p. 1023, 2022. [Online]. Available: \ <https://www.mdpi.com/1593116>
- [6] K. K. Y. Shin, J. Lee, and H. S. Park, “Low-cost automated hydroponic system for urban farming using IoT,” *HardwareX*, vol. 14, e00406, 2023. [Online]. Available:

https://www.researchgate.net/publication/376389154_SMART_GROW_-_Low-cost_automated_hydroponic_system_for_urban_farming

- [7] R. K. Febriana, T. W. Utami, and B. R. Aditya, “Enhancing hydroponic farming productivity through IoT-based monitoring and control,” in Proc. 19th Int. Conf. Web Information Systems and Technologies (WEBIST), 2024. [Online]. Available: <https://www.scitepress.org/Documents/2024/127413/>
- [8] A. A. Sneineh, F. M. Al-Qudah, and R. F. Al-Soud, “Design of a smart hydroponics monitoring system using an ESP32 microcontroller and IoT,” Journal of King Saud University – Computer and Information Sciences, 2023. [Online]. Available: <https://doi.org/10.1016/j.mex.2023.102401>
- [9] E. M. Olalla, F. Torres, and M. D. Rodríguez, “Fuzzy control application to an irrigation system of hydroponic strawberries,” Sensors, vol. 23, no. 8, p. 4123, 2023. [Online]. Available: <https://doi.org/10.3390/s23084088>
- [10] M. A. Awal, R. Rahman, and S. Parvez, “A smart IoT-based hydroponics system for small-scale agriculture,” Internet of Things and Cyber-Physical Systems, vol. 5, pp. 23–34, 2025. [Online]. Available: <https://doi.org/10.1016/j.atech.2025.101163>
- [11] S. K. Berger, L. Martin, and F. Dupont, “Analysis of volatile organic compounds from compost: A two-component sampling strategy,” Atmosphere, vol. 16, no. 5, p. 591, 2025. [Online]. Available: <https://www.mdpi.com/2073-4433/16/5/591>

- [12] M. Szyłak-Szydłowski, P. Nowak, and M. Zieliński, “Assessment of odour emission during the composting process using multi-instrumental methods,” *Materials*, vol. 18, no. 2, p. 832, 2025. [Online]. Available: <https://www.mdpi.com/3315156>
- [13] M. Rivai, A. T. Saputra, and A. H. Kridalaksana, “Implementation of Real-Time Fuzzy Logic Control for NFT-Based Hydroponic System in an IoT Environment,” in *Proc. 2016 6th Int. Conf. System Engineering and Technology (ICSET)*, Bandung, Indonesia, 2016, pp. 130–135. doi: 10.1109/ICSEngT.2016.7849641. [Online]. Available: <https://doi.org/10.1109/ICSEngT.2016.7849641>
- [14] T. A. Kumar, R. Venkatesan, and A. A. Venkatesh, “Smart Sprouts: Fuzzy Logic–Driven IoT-Based Smart Water Management System for Vertical Farming,” in *Proc. 2023 Int. Conf. Smart Electronics and Communication (ICOSEC)*, Trichy, India, 2023, pp. 243–248. doi: 10.1109/ICOSEC57815.2023.10352963. [Online]. Available: <https://doi.org/10.1109/ICOSEC58147.2023.10275927>
- [15] M. A. Rosli, N. A. A. Majid, and M. S. A. Bakar, “Design of an IoT-Based Smart Irrigation and Fertilization System Using Fuzzy Logic for Chili Plant,” in *Proc. 2020 IEEE Int. Conf. Automatic Control and Intelligent Systems (I2CACIS)*, Shah Alam, Malaysia, 2020, pp. 158–163. doi: 10.1109/I2CACIS49202.2020.9238554. [Online]. Available: <https://doi.org/10.1109/I2CACIS49202.2020.9140199>
- [16] J. C. V. Puno, J. J. I. Haban, J. D. Alejandrino, A. A. Bandala, and E. P. Dadios, “Design of a nutrient film technique hydroponics system with fuzzy logic control,” in *Proc. IEEE TENCON*, Osaka, Japan, Nov. 2020, pp. 403–409. <https://doi.org/10.1109/TENCON50793.2020.9293749>

- [17] S. B. Triantino, A. B. Utomo, A. Mulwinda, N. A. Salim, A. Hangga, and A. M. Nisa, "Control system of nutrient solution pH using fuzzy logic for hydroponics system," in Proc. 9th IEEE ICITACEE, Semarang, Indonesia, Aug. 2022, pp. 71–77.
<https://doi.org/10.1109/ICITACEE55701.2022.9924108>
- [18] P. Gourshettiwar and K. T. V. Reddy, "Development of an IoT-enabled real-time monitoring and alert system model for hydroponic farming," in Proc. 5th IEEE ICTMIM, 2025, pp. 1161–1168.
<https://doi.org/10.1109/ICTMIM65579.2025.10988398>
- [19] K. M. C. Dalogdog, A. C. Elmido, E. A. Cardoso, R. C. Baiño Jr., E. D. M. Urgel, J. E. Cartel, R. T. Reas, A. R. T. Casio, D. C. de Paz, A. C. España, and A. B. Culaba, "Effects of organic nutrient solutions on the quality and yield of hydroponically grown vegetable," in Proc. 15th IEEE HNICEM, 2023, pp. 1–5.
<https://doi.org/10.1109/HNICEM60674.2023.10589073>
- [20] J. B. N. G. Olmedo, C. J. O. Costelo, D. A. Y. Brigaste, E. A. Cardoso, R. C. Baiño Jr., M. L. A. Dollete, A. B. Palmes, R. T. Reas, D. C. de Paz, A. C. España, and A. B. Culaba, "The performance of organic nutrient solutions in hydroponic vegetable production: Inputs for process quality control," in Proc. 15th IEEE HNICEM, 2023, pp. 1–6.
<https://doi.org/10.1109/HNICEM60674.2023.10589074>